1 Score each answer two ways

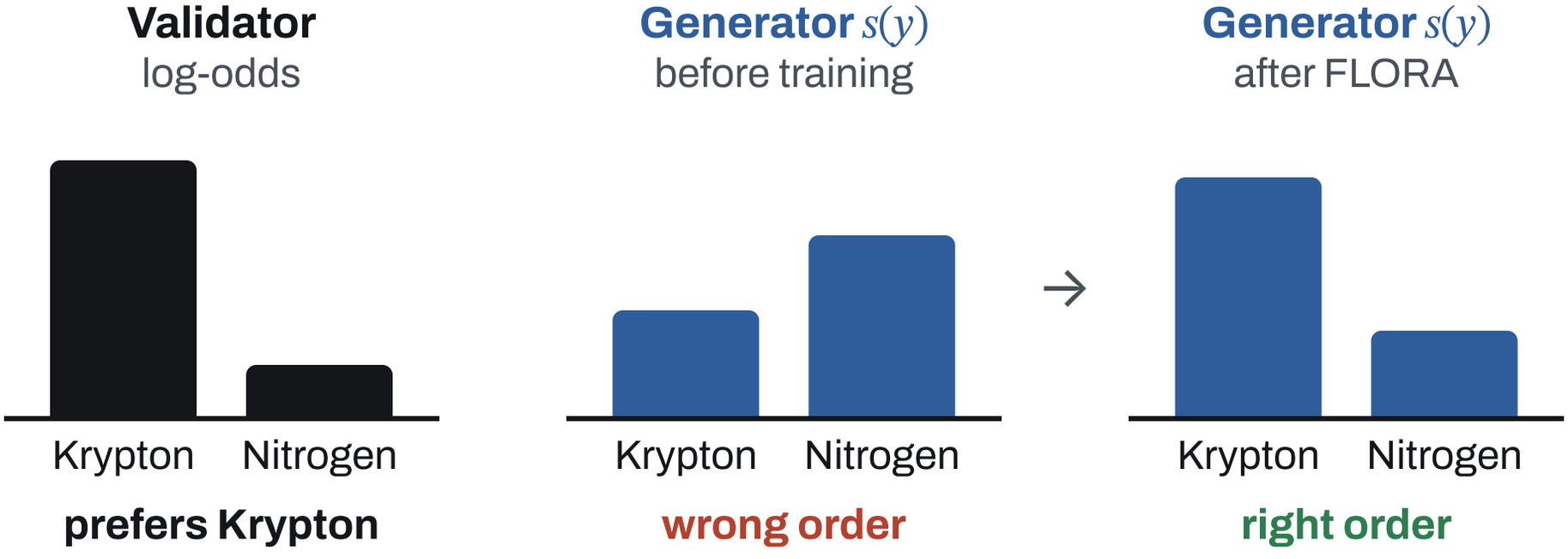
Corrected generator score

$$s(y) = \log p_{\mathrm{G}}(y \mid x) - \log p_{\mathrm{G}'}(y \mid x)$$

Validator log-odds
from “Is y correct?”

2 Pair answers; the validator picks the winner

“Example of a noble gas?” **Krypton** (correct, rare) vs **Nitrogen** (wrong)



3 Train with a ranking loss

$$\mathcal{L} = -\log \sigma(\, s(y^{+}) - s(y^{-})\,)$$

+ likelihood terms on labelled answers

y^{+}: the answer the validator prefers; training raises its score above y^{-}’s. Bars are illustrative.